

Guardrails for the Record Race: What a CsPbI₃ Cell Can Honestly Claim (23.8% → 20.8%)

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Abstract: Simulation records for CsPbI₃ solar cells now exceed the certified experimental frontier (22.02%), but the numbers rest on undisclosed assumptions. We screen an ITO/TiO₂/CsPbI₃/CBTS/Ni cell in SCAPS-1D over a stated box—absorber 0.4–2.2 μm , N_A 10^{15} – $3 \times 10^{16} \text{ cm}^{-3}$, N_t 10^{12} – 10^{15} cm^{-3} , ETL N_D 10^{16} – 10^{19} cm^{-3} , E_g 1.65–1.75 eV with linked affinity, interfaces 10^8 – 10^{12} cm^{-2} —and price every assumption with computed sweeps. At an attainable gap ($E_g = 1.68 \text{ eV}$) the model gives 22.70%, falling to 22.12% at $R_s = 1 \Omega\text{cm}^2$ ($E_g = 1.70 \text{ eV}$); these derated numbers, not the box best, are the honest comparators. The best point reaches 23.84% (V_{oc} 1.250 V, J_{sc} 21.19 mA/cm^2 , FF 89.97%) at 2.2 μm , $N_t = 10^{12} \text{ cm}^{-3}$, $R_s = 0$, 10^{10} cm^{-2} interfaces and a small ETL-side electron spike ($\chi_{\text{TiO}_2} = 3.8 \text{ eV}$, adopted after a sweep showed +0.33 points). At routine settings the same model gives **20.82%** (ITO 4.0 eV, $N_t = 10^{15} \text{ cm}^{-3}$, 1.0 μm , 10^{10} cm^{-2} interfaces). An interface dose–response on our own cell runs 25.11% to 21.20%; the published 24.17% and 24.24% values fall below our no-interface band and are not mapped onto our curve. Scripted parasitic sweeps give 23.43% at $R_s = 1 \Omega\text{cm}^2$. A Morris screen (56 runs, bootstrap 95% CIs) gives 1.02 points/decade for interface quality [0.84, 1.22] against 0.68 for bulk defects [0.35, 0.99]—intervals that overlap, so the two are not statistically separated; repeating the dose–response at routine defects makes the interface response nearly flat, so the ranking is regime-dependent. All definition files, sweep receipts, and raw curves are archived in the public repository for independent reproduction.

Keywords: CsPbI₃; perovskite solar cell; SCAPS-1D; CBTS; reproducibility; device simulation

1.0 Introduction

All-inorganic CsPbI₃ perovskite solar cells combine the thermal stability lacking in hybrid organic–inorganic perovskites with a bandgap near 1.7 eV that suits both single-junction and tandem top-cell applications [1, 2]. After years below 19%, certified laboratory efficiencies have recently passed 22%: 22.02% certified (22.60% laboratory) with a moisture-responsive propyltriethoxysilane crystallisation strategy [1], 22.05% by fluorination-assisted interfacial dipole engineering [2], 21.71% with dual-interface and bulk co-passivation at an open-circuit voltage of 1.27 V [3], and 21.43% certified for inverted architectures [4]. Each record was won at the interfaces and in the bulk defect chemistry, which motivates a simulation study that treats those exact levers honestly.

The one-dimensional solar cell capacitance simulator (SCAPS-1D) is the workhorse for such pre-experimental screening, and the ITO/TiO₂/CsPbI₃/CBTS (copper barium thiostannate) stack of Hossain et al. [5] is the most-prescribed CsPbI₃ baseline, progressing from 17.90% over 96 screened transport-layer combinations [6] to 19.06% after optimisation. Subsequent simulation studies report up to 24.17% on the same stack [7] and 24.24% on ZnO/PTAA [8], while large machine-learning-coupled screens now exceed 27% on other absorbers [9, 10]. Three problems accompany this inflation. First, nearly every study optimises one variable at a time (OVAT), so parameter interactions—notably absorber thickness versus defect-limited diffusion length—are invisible by construction. Second, the highest numbers are obtained with interface recombination removed (stated explicitly in [8]) or with defect densities far below experimental reach, and raw data are routinely withheld [5]—we

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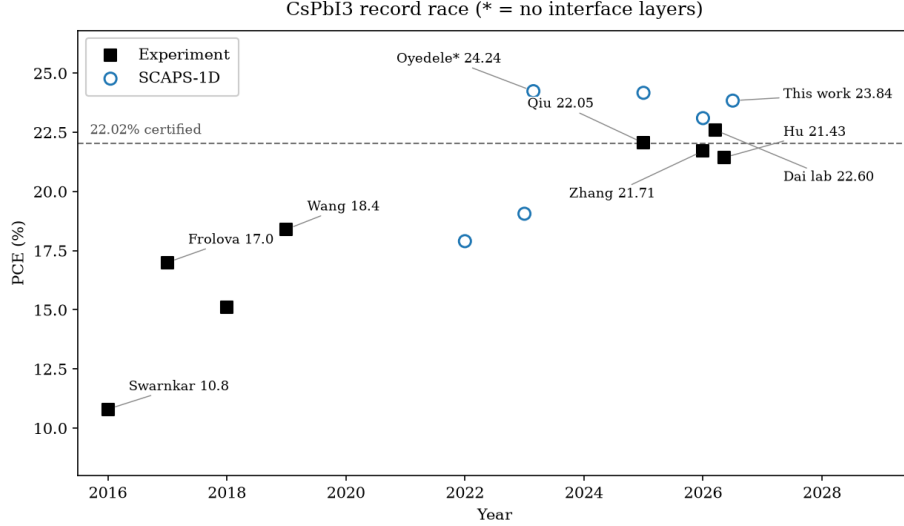


Figure 1: CsPbI3 record race. Asterisks mark simulation studies without interface layers.

report our own data the same critical way, and where we could not reproduce others for lack of published parameters, we say so instead of inferring inflation. Third, scripted sweeps mix unit systems: SCAPS definition files store SI units while script ‘set’ commands expect practical units, so SI-valued inputs silently inject a $10^6 \times$ overdose and fabricate ghost cells (scripting conventions are documented in Appendix A).

This work contributes (i) a five-factor grid screen (96 configurations) plus refinement (36 runs) of the ITO/TiO₂/CsPbI₃/CBTS/Ni cell under interface-inclusive guardrails, with a post-hoc ETL-affinity adoption ($\chi_{\text{TiO}_2} = 3.8$ eV, +0.33 points by sweep) giving a best point of **23.84%** in the stated box; (ii) an interface dose–response computed on our own cell; (iii) scripted (not analytic) series- and shunt-resistance accounting; and (iv) a Morris sensitivity screen with bootstrap intervals, reported with explicit range-dependence. Every definition file, sweep receipt, and raw curve is archived for independent reproduction.

2.0 Literature: a decade of CsPbI3 progress

Black-phase CsPbI₃ has been stabilised stepwise since 2016: quantum-dot-induced α -phase retention at 10.8% [11], co-evaporated planar cells at 17.0% [12], thermodynamically stable γ -films [13], graded-gap 14.4% devices [14], and compositionally engineered 13–14% cells [15, 16], all on TiO₂ with Spiro-OMeTAD, PTAA or carbon contacts [17–19]. The 2019 aziridinium result at 18.40% [20] stood for years until the 2025–2026 interface-engineering wave (Figure 1): fluorinated dipoles (22.05% [2]), dual-interface plus bulk co-passivation with Voc 1.27 V (21.71% [3]), moisture-responsive crystallisation (22.60% laboratory, 22.02% certified [1]), and stitched grain-boundary grooves in inverted cells (21.43% certified [4]). The certified frontier is therefore 22.02% (Dai; laboratory best 22.60%; Qiu’s 22.05% is uncertified), and every recent jump came from defect and interface control—the precise levers a simulation study must treat honestly.

Simulation studies trace a parallel inflation curve (Table 1). The CBTS screening programme reached 17.90% over 96 transport-layer combinations [6] and 19.06% after full optimisation [5], the baseline we reproduce. Later claims diverge upward as realism constraints are relaxed: 23.10% with ZnO/Spiro in one report [21], 24.17% on our exact stack [7], and 24.24% on ZnO/PTAA without interface layers (stated) [8]. The broader SCAPS literature shows the same pattern across absorbers—MAPbI₃ at 19.30% with inorganic contacts [22], gradient-doped FAPbI₃ at 20.34% [23], a 28.38% bifacial triple-cation cell [10], and an explicitly idealised 40.17% MASnI₃ result that exceeds the

Table 1: Selected CsPbI₃ results: experiment versus simulation (IF = interface layers).

Study	Stack	Type	PCE (%)	IF
Swarnkar 2016	TiO ₂ /Spiro/MoOx	Exp.	10.8	–
Frolova 2017	TiO ₂ /PTAA	Exp.	17.0	–
Wang 2019	TiO ₂ /Spiro	Exp.	18.4	–
Qiu 2025	DFAz dipole	Exp.	22.05	passivated
Dai 2026	TiO ₂ /PTABr/Spiro	Exp.	22.60/22.02	passivated
Hu 2026 (Joule)	inverted HPDA	Exp.	21.43 cert.	2D-stitched
Hossain 2022	TiO ₂ /CBTS	Sim.	17.90	on
Hossain 2023	TiO ₂ /CBTS/Ni	Sim.	19.06	on
Ristiansyah 2026	ZnO/Spiro	Sim.	23.10	on
Nazli 2025	TiO ₂ /CBTS/Ni	Sim.	24.17	unclear
Oyedele 2023	ZnO/PTAA	Sim.	24.24	OFF
This work	TiO ₂ /CBTS/Ni	Sim.	23.84	on

Shockley–Queisser bound [24, 25]. Section 7.0 proposes the common reporting standard these numbers would need—interface status with dose level, defect floors with cross-sections—and demonstrates it on our own cell, since the published parameter tables are too incomplete to re-place anyone else’s.

3.0 Simulation framework and guardrails

Devices were simulated in SCAPS-1D v3.3.10 under AM1.5G illumination (100 mW/cm²) at 300 K unless stated. The baseline reproduces Hossain et al. [5]: ITO front contact (4.0 eV), 30 nm TiO₂ electron transport layer (donor density 9×10^{16} cm⁻³), CsPbI₃ absorber (800 nm initial, $E_g = 1.694$ eV, $\chi = 3.95$ eV, $N_A = 10^{15}$ cm⁻³, $N_t = 10^{15}$ cm⁻³), 100 nm CBTS hole transport layer ($N_A = 10^{18}$ cm⁻³), and Ni back contact (5.5 eV). Crucially, neutral interface recombination blocks ($N = 10^{10}$ cm⁻², $\sigma = 10^{-19}$ m²) are present at both heterojunctions, and the absorber uses the coefficients-A-and-B absorption model ($A = 10^7$ m⁻¹eV^{-1/2}) so that bandgap sweeps act on the absorption edge rather than a frozen file spectrum. Our unmodified baseline yields 18.99% (Voc 1.122 V, Jsc 19.68 mA/cm², FF 86.00%), reproducing the published 17.9–19.06% band and validating the harness.

Guardrails enforced throughout: bulk defect floor 10^{12} cm⁻³ (single-crystal grade; nothing lower), absorber doping ceiling 3×10^{16} cm⁻³, ETL affinity adopted at $\chi_{\text{TiO}_2} = 3.8$ eV after a sweep (3.8–4.2 eV) showed the surveyed 4.0 eV is not optimal, conduction-band offset held near zero by linking χ to E_g ($\chi = 3.95 + 0.5(E_g - 1.694)$), and automatic rejection of runs with Voc above $E_g/q - 0.28$ V (~ 1.37 V at 1.65 eV, a plausibility ceiling; the model sets $k_{\text{rad}} = 0$) or FF above Green’s ideal-diode value at the computed Voc (90.1% at 300 K for the champion) as non-physical. Series and shunt resistances are scriptable in SCAPS 3.3.10 (set `external.Rs/Rsh`; `scriptdescription3310.txt`), so this section reports computed parasitic sweeps. Sweeps ran in parallel on four workers with checkpointed receipts; of 132 grid configurations, 126 converged and enter the analysis (6 non-converged worker runs excluded and listed in the receipts: three Round-A runs, all at $N_A = 10^{15}$ cm⁻³ off the champion corner, and three Round-B runs, all at 1.8 μm thickness). A duplicate champion run reproduces the coarse-grid value exactly, confirming determinism; a confirmatory run at finer IV step gives 23.8365% (FF 89.97%), reported as 23.84% throughout. Absorber densities of states are capped at $N_c = 1.1 \times 10^{19}/N_v = 8.2 \times 10^{19}$ cm⁻³, ten times below the Table-1 print (10^{20} cm⁻³, above solid density and hence unphysical).

SCAPS-1D solves the Poisson equation coupled to electron and hole continuity with drift-

Table 2: Layer input parameters (script units; *unmodified baseline* values as reproduced from Hossain et al. [5]: $\chi_{\text{TiO}_2} = 4.0$ eV, $N_{D,\text{ETL}} = 9 \times 10^{16} \text{ cm}^{-3}$. Screening moves to $N_D = 9 \times 10^{17}$ and adopts $\chi_{\text{TiO}_2} = 3.8$ eV; those values appear in Table 5, not here). Mobilities in cm^2/Vs ; $v_{th} = 10^5$ m/s all layers; contacts $S_n = S_p = 10^5$ m/s with Ni 5.5 eV / ITO 4.0 eV; front reflection filter 10%, back transmission filter 80%; radiative (K_{rad}) and Auger recombination off.

Layer	d	E_g	χ	ϵ_r	Doping	N_t	μ_n/μ_p	N_c/N_v ; defect $\sigma_n/\sigma_p, E_t$
CBTS HTL	100 nm	1.9	3.6	5.4	N_A 1e18	1e15	30/30	2.2e18/1.8e19; 1e-20/1e-19, 1.5 eV
CsPbI3	800 nm	1.694	3.95	6.0	N_A 1e15	1e15	25/25	1.1e19/8.2e19; 1e-19/1e-19, 0.6 eV
TiO2 ETL	30 nm	3.2	4.0	9.0	N_D 9e16	1e15	20/20	2.0e18/1.8e19; 1e-19/1e-18, 1.6 eV

Table 3: Interface defect layers (both heterojunctions): neutral single level, $\sigma = 10^{-19} \text{ m}^2$ all sides, $E_t = 0.6$ eV.

Interface	$N (\text{cm}^{-2})$	Sweep range (this work)
CBTS/CsPbI3	1e10	10^8 – 10^{12} (+ ETL/HTL-only, $\sigma \times 10$)
CsPbI3/TiO2	1e10	10^8 – 10^{12} (+ ETL/HTL-only, $\sigma \times 10$)

diffusion transport [26, 27]:

$$-\frac{\partial}{\partial x} \left(\epsilon(x) \frac{\partial V}{\partial x} \right) = q[p - n + N_D^+ - N_A^- + p_t - n_t], \quad (1)$$

$$\frac{\partial n}{\partial t} = \frac{1}{q} \frac{\partial J_n}{\partial x} + G_n - R_n, \quad \frac{\partial p}{\partial t} = -\frac{1}{q} \frac{\partial J_p}{\partial x} + G_p - R_p, \quad (2)$$

with Shockley–Read–Hall recombination through the specified bulk and interface defect levels. Optical generation uses the Eg-sqrt (Tauc) model $\alpha = A\sqrt{h\nu - E_g}$, so bandgap sweeps move the absorption edge self-consistently. Full input parameters are listed in Tables 2 and 3; transport-layer screenings that motivate the TiO2/CBTS choice are reviewed in [6, 28].

4.0 Grid screen and adopted champion

Round A explored the full cross-product of absorber thickness (1.0–2.0 μm), acceptor density (10^{15} – 10^{16} cm^{-3}), bulk defect density (10^{12} – 10^{14} cm^{-3}), ETL donor density (9×10^{16} – $9 \times 10^{17} \text{ cm}^{-3}$), and bandgap (1.694/1.65 eV with linked affinity)—96 configurations in a single parallel batch, rather than sequential OVAT slices. The winner (2.0 μm , $N_A = 10^{16} \text{ cm}^{-3}$, $N_t = 10^{12} \text{ cm}^{-3}$, $N_{D,\text{ETL}} = 9 \times 10^{17} \text{ cm}^{-3}$, $E_g = 1.65$ eV) reached 23.47%. Full parameters are collected in Table 5. Round B refined thickness (1.8–2.2 μm), gap (1.65–1.72 eV), and doping around this point (36 configurations) and reached 23.50% (2.2 μm , $N_A = 3 \times 10^{16} \text{ cm}^{-3}$, $N_t = 10^{12} \text{ cm}^{-3}$, $E_g = 1.65$ eV, $\chi = 3.928$ eV). A

Table 4: Guardrail values and what anchors them. “Assumption” means a modelling choice with no direct measurement behind it.

Guardrail	Value	Anchor
Bulk defect floor	$N_t = 10^{12} \text{ cm}^{-3}$	assumption (single-crystal grade)
Interface density	$N_{if} = 10^{10} \text{ cm}^{-2}$	assumption; $S = N\sigma v \approx 100 \text{ cm/s}$ (demanding end of passivity)
Capture cross-section	$\sigma = 10^{-19} \text{ m}^2$	assumption (SCAPS-typical; swept $\times 0.1$ – $\times 10$ here)
Trap level	$E_t = 0.6$ eV mid-gap	assumption (swept 0.3–0.9 eV: ≤ 0.01 -point effect)
V_{oc} ceiling	$E_g/q - 0.28$ V	plausibility screen, not a computed limit ($k_{rad} = 0$)
FF ceiling	Green’s ideal at measured V_{oc}	analytic bound
R_s, R_{sh} basis	scripted sweeps, reported	computed, not assumed zero in headlines

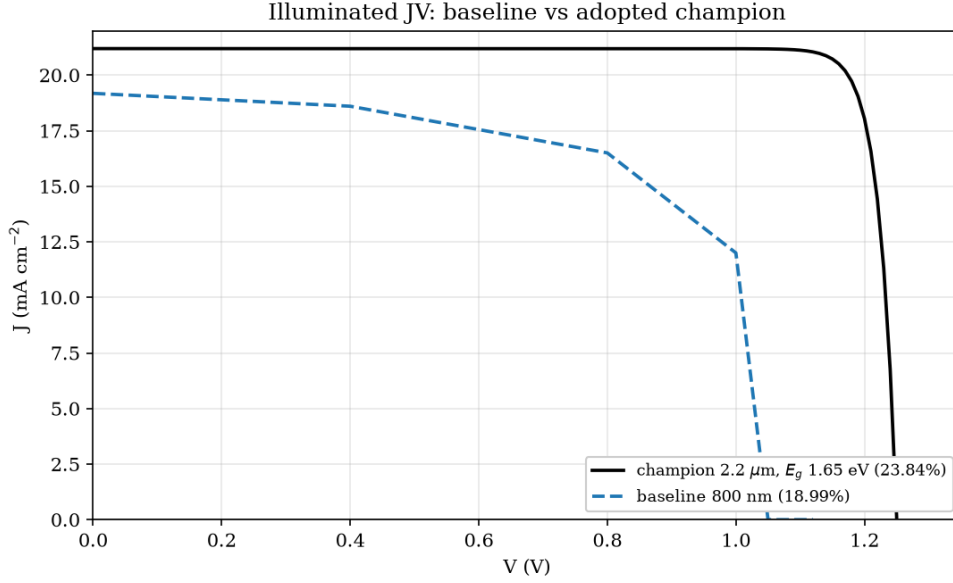


Figure 2: Illuminated current–voltage curves: baseline (800 nm, 18.99%) versus adopted champion (2.2 μm , 23.84%).

subsequent ETL-affinity sweep showed the surveyed 4.0 eV is not optimal: the surveyed- χ Round-B best is 23.50% (B030 = 23.5041), 3.9 eV gives 23.80%, 3.8 eV gives **23.84%** (V_{oc} 1.250 V, J_{sc} 21.19 mA/cm^2 , FF 89.97%; Figure 2), adopted as the box best (+0.33 points), while 4.1 and 4.2 eV collapse to 21.92% and 19.87%. Two features of this point matter. First, it sits at the corner of the allowed box: N_A at its ceiling, N_t at its floor, $N_{D,ETL}$ at the top of its tested range, E_g at its bottom, thickness at the top of the refined range. Second, the best grid point at the literature gap $E_g = 1.694$ eV reaches only 22.51%, so roughly one point comes from the gap choice alone. Every top-ten configuration carries $E_g = 1.65$ eV, gaining ~ 1.5 mA/cm^2 over 1.694 eV; the ETL side now presents a small +0.13 eV electron spike that raises the interface recombination activation energy (Section 5.0).

From diffusion estimates ($\mu_n = 25$ cm^2/Vs , $\sigma = 10^{-19}$ m^2): $L \approx 80$ μm at 10^{12} cm^{-3} , ≈ 8 μm at 10^{14} cm^{-3} , ≈ 2.5 μm at 10^{15} cm^{-3} . The map (Figure 3) is therefore nearly thickness-insensitive at 10^{14} cm^{-3} (near-vertical contours)—the oft-quoted “collapse below 1.5 μm ” cannot be diffusion-limited and is withdrawn. The genuine thickness–defect interaction appears at 10^{15} cm^{-3} , where a dedicated sweep gives 20.06% (0.5 μm), 20.72% (0.8 μm), 20.82% (1.0 μm), 20.77% (1.5 μm): thickness matters exactly where L approaches the film, which single-variable sweeps at fixed defect levels cannot resolve. ETL donor density confirms the literature optimum (9×10^{17} cm^{-3}) and transport-layer thicknesses are insensitive (100/30 nm retained), both consistent with [5]. The current–voltage curve (Figure 2) is square and artefact-free, with a V_{oc} deficit of 0.40 V against E_g/q .

The 4.85-point stacked gain decomposes along the screening path (Table 5 parameters in sequence, illustrative ordering only): thickness 800 nm to 1.5 μm adds 0.19 points (19.18%), acceptor doping to 10^{16} cm^{-3} adds 0.93 points (20.11%), bulk-defect reduction to 10^{12} cm^{-3} adds ~ 1.6 points ($\sim 21.7\%$), the joint E_g – χ move to 1.65 eV with ETL donor optimisation contributes ~ 1.8 points (4.50 to the surveyed best), and the adopted χ_{TiO_2} adds the final 0.33 points (modulo rounding). This sequential attribution is order-dependent and illustrative, not causal. Open-circuit voltage climbs 1.122 to 1.250 V while the fill factor rises 86.0 to 90.0%, the signature of SRH-lifetime recovery rather than band-shift artefacts.

Round A’s top ten (Table 6) all carry $E_g = 1.65$ eV; the top seven sit at $N_t = 10^{12}$ cm^{-3} while rows 8–10 at $N_t = 10^{13}$ cm^{-3} trail by ~ 0.4 points through V_{oc} loss (1.233 to 1.221 V). Round B refined thickness (1.8–2.2 μm), gap (1.65–1.72 eV) and doping (36 runs) and moved the best from

Table 5: Champion device parameters (script units) and result.

Layer	d	E_g (eV)	χ (eV)	Doping (cm^{-3})	N_t (cm^{-3})
CBTS HTL	100 nm	1.9	3.6	NA 1e18	1e15
CsPbI3 absorber	2.2 μm	1.65	3.928	NA 3e16	1e12
TiO2 ETL	30 nm	3.2	3.8 (adopted; surveyed 4.0)	ND 9e17	1e15
Interfaces (both)	–	–	–	–	1e10 cm^{-2}
Result	Voc 1.250 V, Jsc 21.19 mA/ cm^2 , FF 89.97%, PCE 23.84%				

Table 6: Round A leaderboard (best ten of 96; $N_{D,ETL} = 9 \times 10^{17} \text{ cm}^{-3}$; $\chi_{\text{TiO}_2} = 4.0 \text{ eV}$ surveyed throughout).

d (μm)	N_A	N_t	E_g	Voc	Jsc	PCE
2.0	1e16	1e12	1.65	1.233	21.18	23.47
1.8	1e16	1e12	1.65	1.233	21.16	23.45
1.5	1e16	1e12	1.65	1.234	21.11	23.40
1.5	1e15	1e12	1.65	1.233	21.11	23.29
2.0	1e15	1e12	1.65	1.233	21.18	23.29
1.0	1e16	1e12	1.65	1.234	20.90	23.18
1.0	1e15	1e12	1.65	1.233	20.90	23.11
1.8	1e16	1e13	1.65	1.221	21.16	23.04
2.0	1e16	1e13	1.65	1.219	21.18	23.03
1.5	1e16	1e13	1.65	1.223	21.11	23.03

23.47% to 23.50% (coarse grid), i.e. the landscape is converged rather than climbing.

Round B's refinement is documented in full because its plateau is itself evidence: the best ten span only 23.32–23.50% (Table 7), with thickness 1.8–2.2 μm and N_A (1–3) $\times 10^{16} \text{ cm}^{-3}$ numerically indistinguishable at grid resolution once $N_t = 10^{12} \text{ cm}^{-3}$ and $E_g = 1.65 \text{ eV}$ are fixed (the ten span 23.32–23.50%, a 0.18-point range against a 0.017-point coarse/fine IV-step uncertainty, so the leading cluster cannot be separated even though the 10th entry trails observably). The $N_t = 3 \times 10^{12} \text{ cm}^{-3}$ entries trail by ~ 0.05 points, confirming the defect floor binds the optimum.

5.0 Band alignment

Figure 4 constructs the equilibrium alignment from the adopted champion parameters. The absorber conduction edge (-3.928 eV) lies 0.13 eV *below* the TiO2 edge (-3.80 eV), so electrons face a small

Table 7: Round B leaderboard (best ten of 36; $N_{D,ETL} = 9 \times 10^{17} \text{ cm}^{-3}$).

d (μm)	N_A	N_t	E_g	PCE
2.2	3e16	1e12	1.65	23.50
2.0	3e16	1e12	1.65	23.50
2.2	1e16	1e12	1.65	23.48
2.0	1e16	1e12	1.65	23.47
1.8	3e16	1e12	1.65	23.47
2.2	3e16	3e12	1.65	23.45
1.8	1e16	1e12	1.65	23.45
2.0	3e16	3e12	1.65	23.44
1.8	3e16	3e12	1.65	23.42
2.2	1e16	3e12	1.65	23.32

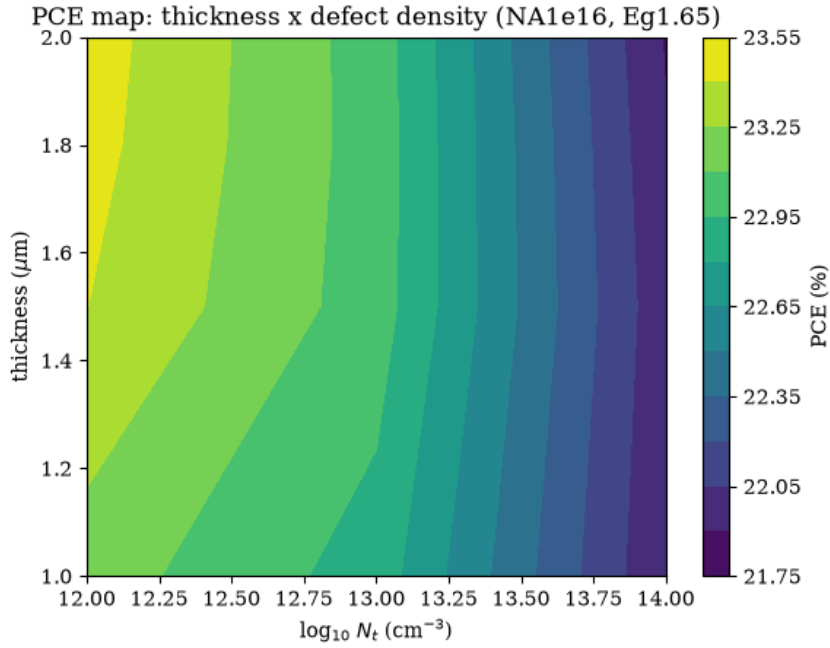


Figure 3: Efficiency map over absorber thickness and bulk defect density (slice at $N_A = 10^{16} \text{ cm}^{-3}$, $E_g = 1.65 \text{ eV}$).

spike entering the ETL. The data support reading this as recombination physics, not transport: the interface recombination activation energy $E_a = E_{c,\text{abs}} - E_{v,\text{HTL}}$ grows from 1.37 eV at $\chi_{\text{CBTS}} = 3.4$ to 1.77 eV at 3.8 eV, and efficiency rises monotonically through zero hole-side offset (20.22%, 22.07%, 23.77%, 23.81%)—a transport reading predicts the barrier end should be worst, and it is the best. On the hole side, the absorber valence edge (-5.578 eV) lies 0.08 eV *below* the CBTS edge (-5.50 eV), a shallow hole-favourable offset on the wrong side of the ideal -0.3 to -0.1 eV window (-3.928 vs -3.80 ; -5.578 vs -5.50). The same activation logic applies to the ETL side ($E_a = 1.58 \text{ eV}$ at surveyed alignment), which is why the spike buys rather than costs efficiency. The adopted $\chi_{\text{TiO}_2} = 3.8 \text{ eV}$ sits below the surveyed 4.0 eV and is reported as a post-hoc adoption, not a surveyed value.

6.0 Device characterisation

The quantum efficiency (Figure 5) is flat at $\sim 89\text{--}90\%$ from 300 nm with the absorption edge red-shifted from 731 nm (baseline, 1.694 eV) to 752 nm (champion, 1.65 eV), directly explaining the J_{sc} gain. The UV plateau assumes the definition file's flat 10% front reflection with no dispersive parasitic absorption in ITO or TiO_2 , so it is an upper envelope in the blue rather than a prediction; with TiO_2 absorbing above 3.2 eV, of order 1 mA/cm² of the reported J_{sc} is at stake in the UV, which is large next to the points debated here and is stated as an uncertainty, not corrected. Capacitance–voltage analysis at 1 MHz (Figure 6) shows rising capacitance with forward bias (31 to 39 nF/cm² over 0–0.5 V). The low-bias $1/C^2$ branch is linear ($R^2 = 1.000$) with the normal sign for a p-type absorber; its slope gives $N_A = 2.93 \times 10^{16} \text{ cm}^{-3}$, essentially the $3 \times 10^{16} \text{ cm}^{-3}$ input, extrapolating to $V_{bi} = 1.30 \text{ V}$ —a free validation of the doping from the solver's own electrostatics. The SCAPS-extracted width (0.29 to 0.22 μm) sits an order of magnitude below the 2.2 μm absorber throughout (the parallel-plate check gives 0.17 μm at 0 V, consistent in order with extra deep-level capacitance unaccounted). The illuminated band diagram at short circuit (Figure 7) exhibits clean quasi-Fermi splitting, and generation exceeds recombination by at least two orders of magnitude at every mesh point of the full 2.33 μm short-circuit profile (minimum ratio $\sim 3 \times 10^2$ at $x \approx 0.2 \mu\text{m}$ near the back contact, rising above 10^5 at both contacts; Figure 8). All profile statements in this section are at short

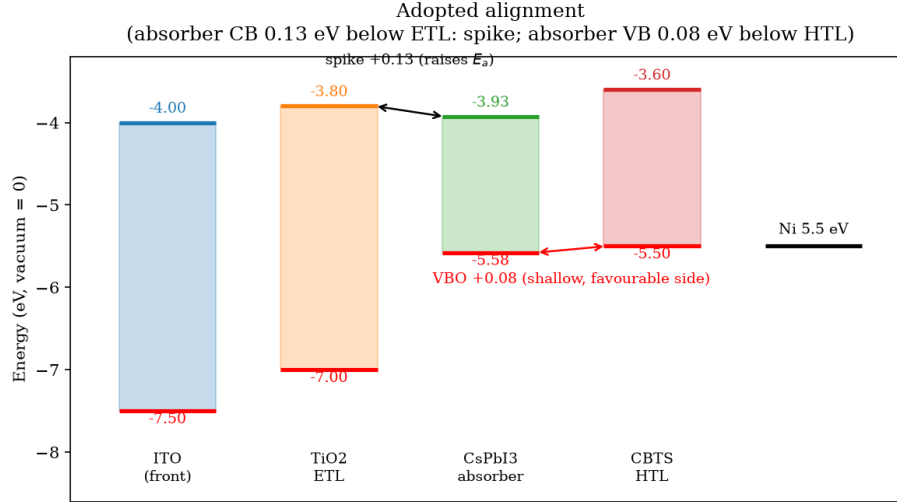


Figure 4: Equilibrium band alignment of the champion stack from input parameters (energies in eV vs vacuum).

Table 8: Temperature dependence at the surveyed base (1 sun), not re-run at the adopted affinity: the 300 K row uses the fine-IV-step confirmatory run (23.49%); all other rows use coarse-grid sweep values.

T (K)	Voc (V)	Jsc (mA/cm ²)	FF (%)	PCE (%)
275	1.271	21.19	90.75	24.44
300	1.234	21.19	89.78	23.49
350	1.157	21.20	87.91	21.57
400	1.078	21.20	85.92	19.64
475	0.955	21.20	82.50	16.70

circuit; nothing here constrains the maximum-power or open-circuit regime.

Temperature dependence (Figure 9, Table 8) follows the established trend [5]: efficiency falls from 24.44% at 275 K to 16.70% at 475 K, driven by Voc (1.271 to 0.955 V) and FF while Jsc stays pinned at ~ 21.19 mA/cm², i.e. $-0.16\%/K$ relative (-0.039 points/K) over 300–400 K. The 275 K fill factor (90.75%) sits *below* Green’s ideal-diode value at that Voc and temperature (90.86%), so no artefact needs invoking; likewise the adopted champion’s 89.97% is within 0.12 points of ideal (90.09%), leaving essentially no FF headroom beyond series resistance. The 475 K end of the sweep ignores the temperature dependence of E_g and black-phase stability, so it is a solver stress test, not a prediction.

6.1 Parasitic-resistance accounting

Series and shunt resistances are scriptable in SCAPS 3.3.10 (set `external.Rs/Rsh`; `scriptdescription3310.txt`). Computed scripted sweeps at the adopted base give 23.64% at $R_s = 0.5 \Omega\text{cm}^2$, **23.43%** at $R_s = 1 \Omega\text{cm}^2$, and 23.00% at $R_s = 2 \Omega\text{cm}^2$ (Table 9), agreeing with the analytic derate to 0.02 points. Shunt paths cost 0.15 points at $R_{sh} = 10^4 \Omega\text{cm}^2$ (23.69%); combined $R_s = 1/R_{sh} = 10^4$ was not re-run at the adopted base and is omitted. The best point therefore survives realistic parasitics, but only just: at $R_s = 2 \Omega\text{cm}^2$ the margin over the 22.02% certified record is under one point.

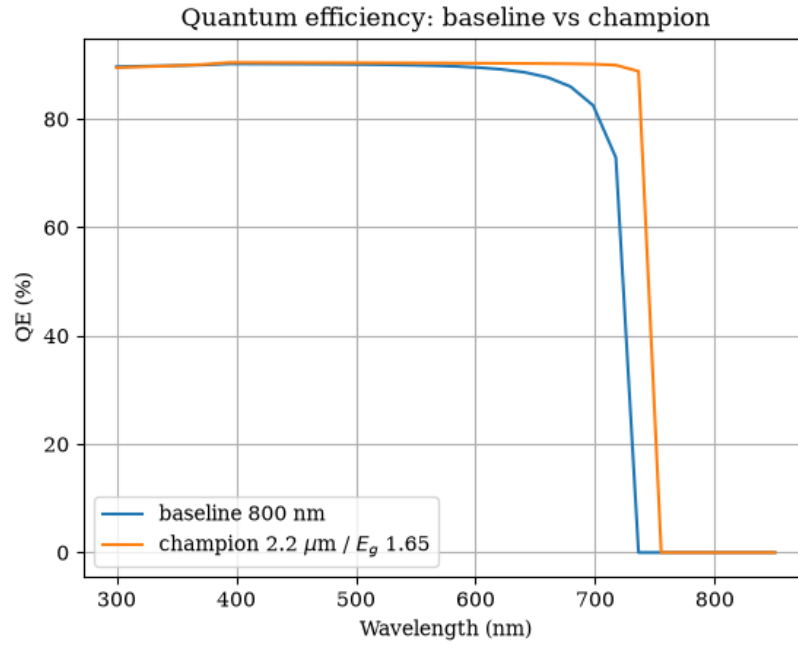


Figure 5: Quantum efficiency of baseline versus champion; the edge shifts 731 to 752 nm.

C-V (1 MHz) and Mott-Schottky

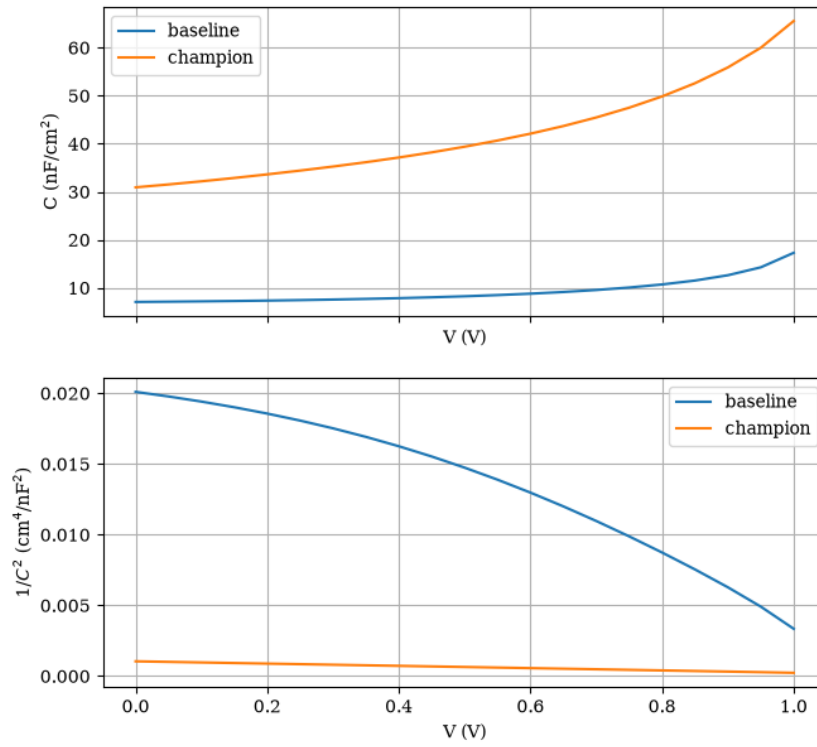


Figure 6: Capacitance–voltage (1 MHz) and Mott–Schottky plots for baseline and champion.

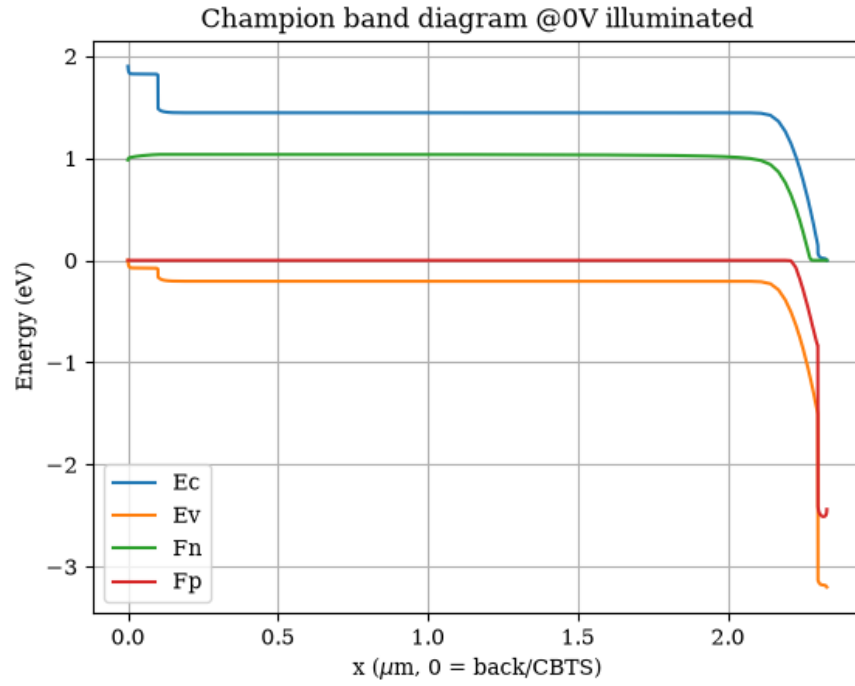


Figure 7: Illuminated band diagram of the champion at short circuit ($x=0$ at the back contact).

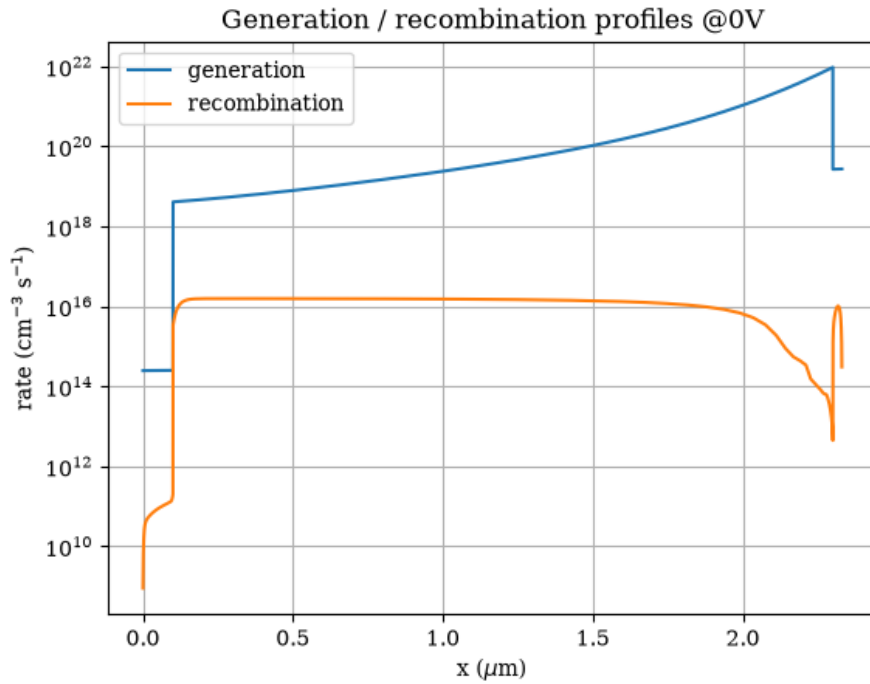


Figure 8: Generation and recombination profiles at short circuit (log scale).

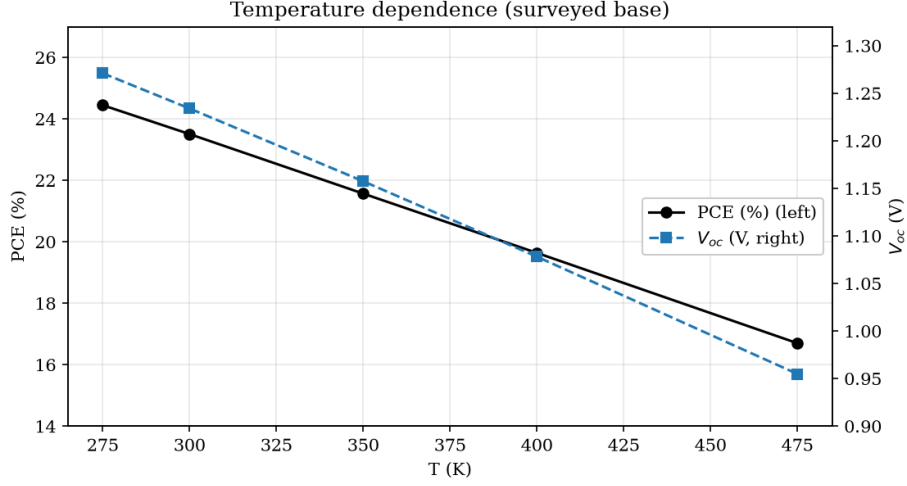


Figure 9: Efficiency and open-circuit voltage versus operating temperature (champion).

7.0 Interface dose–response and the limits of auditing others

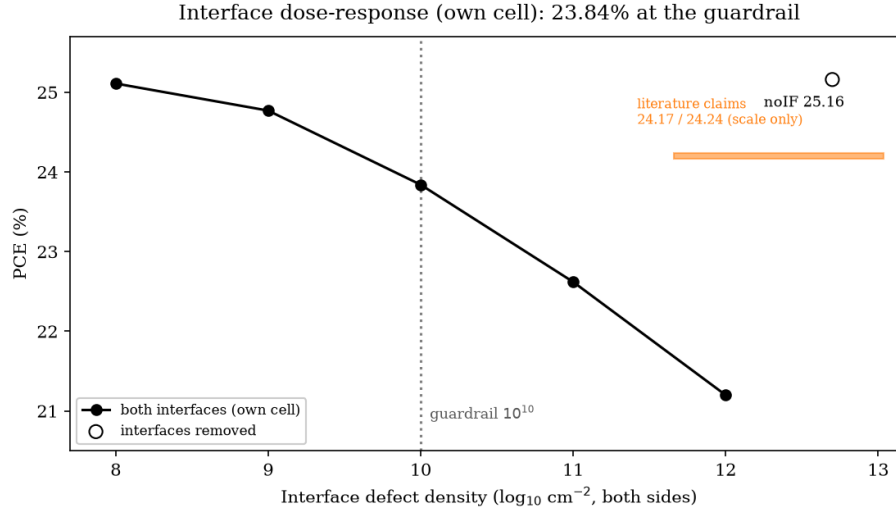
Deleting the interface blocks inflates efficiency by 1.26–1.66 points absolute at the surveyed base (23.40 to 24.87%, 23.47 to 24.73%, 23.50 to 25.16%), almost entirely through a ~ 0.11 V Voc rise as interface recombination vanishes. But the binary ON/OFF comparison understates the audit: the full dose–response at the adopted base (Table 10) runs 25.11% at 10^8 cm $^{-2}$, 24.77% at 10^9 cm $^{-2}$, 23.84% at the 10^{10} cm $^{-2}$ guardrail, 22.62% at 10^{11} cm $^{-2}$, and 21.20% at 10^{12} cm $^{-2}$. The published 24.17% [7] and 24.24% [8] values fall *below* every no-interface value we computed (24.73–25.16% across three surveyed-base cells; 25.16% at the adopted base). Because their stacks, gaps, thicknesses and defect parameters all differ from ours, we do not map their numbers onto our curve; the dose–response is reported strictly as an own-cell result. We could not re-run either study from its published parameters (neither paper prints a complete parameter table), and we state this as a limitation rather than a finding. One no-interface cell (1.5 μ m, Voc 1.382 V) breaches our 1.37 V screening ceiling and is flagged, not trusted.

7.1 Where the missing points go

The inflation is almost entirely a Voc effect (+0.11 V) with Jsc unchanged to within 0.02 mA/cm 2 : removing the 10^{10} cm $^{-2}$ interface states eliminates the dominant SRH channel at the heterojunctions, unpinning the quasi-Fermi splitting. The no-interface Voc values (1.364–1.382 V) reach or slightly exceed our 1.37 V E_g -implied screening ceiling at the 1.5 μ m cell, so those cells are flagged as unphysical rather than celebrated. Fill factor moves non-monotonically (89.8 to 87.0% on the champion) as the recombination profile reshapes. The no-interface 87.0% at Voc 1.364 V sits 3.7 points below Green’s ideal (90.7%): the archived JV curve stays flat to 1.16 V with no kink or S-shape, so there is no visible extraction barrier, but the knee remains SRH-rounded by the surviving bulk channel—removing interfaces unpins Voc without idealising the diode. This bounds the “no extraction barriers” claim to current collection, not to fill factor. Asymmetric tests show both heterojunctions matter: 10^{12} cm $^{-2}$ on the ETL side alone gives 21.35%, on the HTL side alone 21.20%. Raising the capture cross-section $10\times$ at fixed $N_{if} = 10^{10}$ cm $^{-2}$ gives 22.2108%, digit-identical to the $N \times 10$ run at the same base (both 22.2108; receipts in Table 14): for the symmetric single level, rate scales exactly with the $N\sigma$ product. The same identity holds in the bulk: $\sigma \times 10$ at the N_t floor gives 23.2572% (digit-identical to $N_t \times 10$ at the same base, = 23.2572; receipts in Table 14), a 0.25-point cost against the surveyed 23.5041 anchor. (A first attempt via `set ...capture_cross_section` instead restructured the block: applied to one interface it gave 23.8565% (23.8565% on one interface,

Table 9: Best point under computed scripted parasitics (adopted base; all values computed, none derated).

Parasitics	PCE (%)	vs 22.02% cert.	Status
$R_s = 0$ (box maximum)	23.84	+1.82	SCAPS box maximum
$R_s = 0.5 \Omega\text{cm}^2$	23.64	+1.62	realistic good contact
$R_s = 1 \Omega\text{cm}^2$	23.43	+1.41	realistic typical
$R_s = 2 \Omega\text{cm}^2$	23.00	+0.98	margin nearly gone
$R_{sh} = 10^4 \Omega\text{cm}^2$	23.69	+1.67	mild shunt

**Figure 10:** Same cell at graded interface quality: efficiency vs N_{if} (both sides) with the literature 24.17/24.24% scale bar and the no-interface point. The bar is shown for scale only; cross-stack mapping is not claimed.

25.1494% on both)—essentially the no-interface value—so the set-key route neutralises rather than scales, and only the verified definition-file variant is reported; Appendix A documents this second scripting trap and a set-then-get verification rule. Bulk $N_t \times \sigma$ costs 0.25 points at the 10^{12} cm^{-3} floor but 3.12 points at routine 10^{15} cm^{-3} (20.5603% at matched $\sigma \times 1$ geometry down to 17.4433%), while $\sigma \times 0.1$ at the floor gains only 0.04 points (23.53%)—the floor is saturated.) The computed parasitic sweep at the adopted base is tabulated below.

8.0 Morris sensitivity screening

Grid methods with uneven ranges cannot rank physical importance, so we ran a Morris sensitivity screen instead: $r = 8$ trajectories, $p = 4$ levels, $k = 6$ factors (absorber thickness, $\log N_A$, $\log N_t$, $\log N_{D,ETL}$, E_g with the χ link applied deterministically, and $\log N_{if}$ tied on both interfaces), 56 fresh simulations, all converged, spanning 15.6–23.4% (Table 12, Figure 11). Per decade, interface quality leads at 1.02 points (bootstrap 95% CI [0.84, 1.22]) ahead of bulk defect density at 0.68 (CI [0.35, 0.99])—intervals that **overlap**, so with $n = 8$ the two are not statistically separated; per eV, bandgap moves 17.5 points; per μm , thickness 0.72. These per-unit numbers are not commensurable across units, and the interface figure bundles both heterojunctions and depends on the chosen 4-decade range, so the ranking is regime- and range-conditional, not a universal variance share. The decisive test is the same dose–response repeated at routine defects: at $N_t = 10^{15} \text{ cm}^{-3}$ and $1.0 \mu\text{m}$, N_{if} 10^9 – 10^{11} cm^{-2} moves efficiency only from 20.84% to 20.67% (0.16 points over two decades), while the N_t 10^{14} – 10^{15} step costs far more. At the defect floor interfaces dominate; at routine defects bulk defects do. The large σ values and the N_t dose nonlinearity (steepest above 10^{14}) agree. Two

Table 10: Interface dose–response (adopted base unless noted; both sides unless noted).

N_{if} (cm ⁻²)	PCE (%)	Note
10^8	25.11	–
10^9	24.77	–
10^{10} (guardrail)	23.84	this work
10^{11}	22.62	–
10^{12}	21.20	–
10^{12} ETL side only	21.35	surveyed base
10^{12} HTL side only	21.20	surveyed base
10^{10} , $\sigma \times 10$ (verified)	22.21	matches $N \times 10$ (rate $\propto N\sigma$)
removed (noIF)	25.16	near E_g -implied Voc ceiling

Table 11: Guardrail audit: identical cells with interfaces ON versus OFF.

Cell	IF ON (%)	IF OFF (%)	Inflation (pts)
1.5 μm cell	23.40	24.87	+1.47
2.0 μm cell	23.47	24.73	+1.26
Champion (surveyed base)	23.50	25.16	+1.66

disclosures: E_g moves together with χ by construction, and the two interfaces move together. As corroboration without new simulations, a random-forest surrogate trained on all 190 standard-base runs (OOB $R^2 = 0.93$) with Saltelli sampling gives total-order indices $S_T = 0.52$ (interfaces), 0.33 (bulk defects), 0.24 (E_g)—the same ordering as Morris μ^* . Trap depth is a non-factor at this defect level: moving E_t to 0.3 or 0.9 eV changes efficiency by ≤ 0.01 points.

Interface-led sensitivity is consistent with passivation studies in which interface-adjacent defects gate the voltage [29], and with Bayesian optimisation on CsSnI3 where the optimal thickness likewise shifts with defect density [9].

9.0 Discussion

Fabrication plausibility bounds the result, and the computed sweeps price each term. $N_A = 3 \times 10^{16} \text{ cm}^{-3}$ is routinely achievable; $E_g = 1.65 \text{ eV}$ sits below the $\sim 1.7 \text{ eV}$ of the experimental records and contributes roughly one point (best 1.694 eV grid point: 22.51%); $N_t = 10^{12} \text{ cm}^{-3}$ demands near-single-crystal quality ($\tau_{\text{SRH}} \approx 100 \mu\text{s}$). The contacts are load-bearing: Ni 5.5 to 5.0 eV costs 0.23 points, and ITO 4.0 to 4.7 eV collapses the cell to 18.60% (Voc 1.08 V)—SCAPS’s thermionic-only contact boundary, so 4.0 eV is the edge of what this model tolerates rather than a typical ITO value; the routine-defect penalty is larger still, with ITO 4.4 eV costing 0.14 points at surveyed χ (20.82

Table 12: Morris screen design and results ($r = 8$ trajectories, $p = 4$, $\Delta = 2/3$, seed 7; 56/56 converged). μ^* per normalised unit input; last column per physical decade (eV/ μm where noted); $n = 8$ unique directed effects per factor; CI = bootstrap 95% interval (5000 resamples).

Factor	Range	μ^*	σ	n	per decade	95% CI
$\log N_{if}$ (both IF)	10^8 – 10^{12} cm^{-2}	4.08	1.08	8	1.02	[0.84, 1.22]
$\log N_t$ (absorber)	10^{12} – 10^{15} cm^{-3}	2.05	1.40	8	0.68	[0.35, 0.99]
E_g (+ χ link)	1.65–1.75 eV	1.75	0.79	8	17.5/eV	[12.6, 23.2]
Thickness	0.4–2.2 μm	1.29	0.99	8	0.72/ μm	[0.36, 1.07]
$\log N_{D,ETL}$	10^{16} – 10^{19} cm^{-3}	0.68	1.04	8	0.23	[0.02, 0.47]
$\log N_A$ (absorber)	10^{15} – $3 \times 10^{16} \text{ cm}^{-3}$	0.48	0.56	8	0.32	[0.12, 0.58]

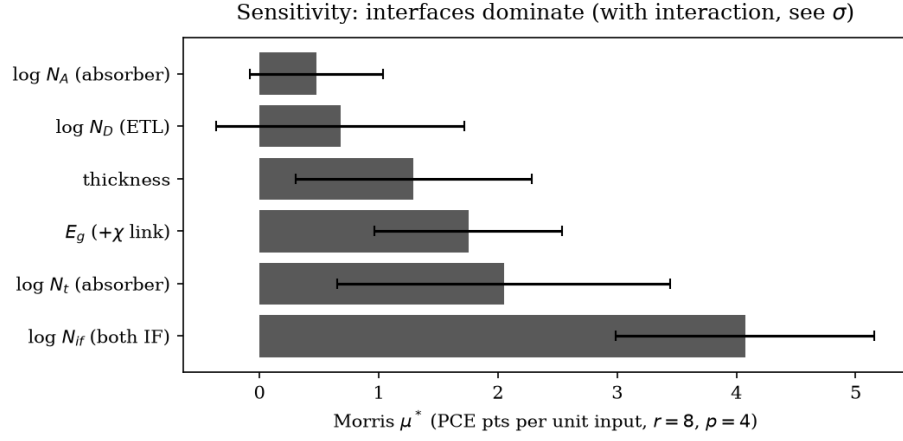


Figure 11: Morris μ^* (bars) with interaction spread σ (whiskers). Ranking is regime-conditional: interfaces lead at the defect floor, bulk defects at routine levels (Table 13); large σ signals interactions.

Table 13: Routine-defect case ($N_t = 10^{15} \text{ cm}^{-3}$, $1.0 \mu\text{m}$): interface dose–response is nearly flat, so bulk defects dominate here. Δ is against the guardrail row of the same base (surveyed: 20.82%; adopted: 20.84%). Values computed; $R_s = 0$ unless stated.

$N_{if} (\text{cm}^{-2})$	PCE (%)	Δ vs guardrail	Base
10^9	20.84	+0.017	surveyed χ
10^{10} (guardrail)	20.82	–	surveyed χ
10^{11}	20.67	–0.147	surveyed χ
10^{10} , ITO 4.4 eV	20.68	–0.14	surveyed χ
10^{10} , ITO 4.4 eV	18.11	–2.73	adopted χ
10^{10} , $R_s = 1 \Omega\text{cm}^2$	–	–	not run

to 20.68%) but 2.73 points at adopted χ (20.84 to 18.11%): the front-contact tolerance window itself shifts with ETL affinity (Table 13). Both affinity sweeps move efficiency through the interface activation energy rather than transport: χ_{CBTS} 3.4 to 3.8 eV gives 20.22 to 23.81%, and χ_{TiO_2} 3.8 to 4.2 eV gives 23.84 to 19.87% (surveyed- χ Round-B anchor 23.50%). At routine settings the model gives 20.82% ($1.0\ \mu\text{m}$, $N_t = 10^{15}\ \text{cm}^{-3}$, surveyed χ), 20.68% with ITO at 4.4 eV on the same base, and 18.11% when that ITO is combined with the adopted χ_{TiO_2} . Against experiment: 23.84% exceeds the 22.02% certified record by 1.82 points and the 22.60% laboratory value by 1.24 points; with computed $R_s = 1\ \Omega\text{cm}^2$ (23.43%) the margins are 1.41 and 0.83 points, and at the attainable gap $E_g = 1.70\ \text{eV}$ the model gives 22.51% ($R_s = 0$, 22.70% at $E_g = 1.68\ \text{eV}$) and 22.12% ($R_s = 1$). The result sits at about three-quarters of the $\sim 30\text{--}31\%$ Shockley–Queisser limit at 1.65 eV.

10.0 Limitations and outlook

A laboratory translation would need, simultaneously: $\sim 1.0\ \mu\text{m}$ CsPbI3 films at routine defect levels for the 20.82% case (time-resolved photoluminescence verified), interface passivation toward $10^{10}\ \text{cm}^{-2}$ following the fluorinated-dipole [2] or dual-interface [3] chemistries, a front contact near 4.0 eV, and $R_s \leq 1\ \Omega\text{cm}^2$ (Table 9). Not demonstrated here: $2.2\ \mu\text{m}$ device-quality CsPbI3 at $N_t \sim 10^{12}\ \text{cm}^{-3}$, and sputtered CBTS directly on CsPbI3 (CBTS typically requires high-temperature sulfurisation), for which we found no precedent—so this section is a requirements list, not a recipe. In-line controls, where applicable: QE edge at 752 nm (Figure 5) and the $-0.16\%/K$ temperature coefficient (Table 8).

Five boundaries delimit this work. First, $N_t = 10^{12}\ \text{cm}^{-3}$ at $2.2\ \mu\text{m}$ demands near-single-crystal films, and with the modelled $\sigma = 10^{-19}\ \text{m}^2$ implies $\tau_{\text{SRH}} \approx 100\ \mu\text{s}$, one to two orders beyond the best reported perovskite films; at routine $10^{15}\ \text{cm}^{-3}$ a dedicated sweep peaks at 20.82% ($1.0\ \mu\text{m}$), 20.68% with ITO at 4.4 eV on the surveyed base, and 18.11% when that ITO meets the adopted χ_{TiO_2} . Second, the contacts (Ni 5.5 eV, ITO 4.0 eV), the adopted $\chi_{\text{TiO}_2} = 3.8\ \text{eV}$, and the shallow off-window hole-side offset (+0.08 eV) are stated-box choices, each priced in Section 9.0. Third, phase stability of black CsPbI3 against the yellow δ -phase, ion migration, $E_g(T)$, and damp-heat degradation are outside SCAPS-1D steady-state scope and must be validated experimentally, following the passivation routes of [1–3]. Fourth, the grid is a screen, not an optimiser: the best point sits at the corner of the allowed box and the leading cluster is numerically inseparable. Fifth, the single-junction ceiling ($\sim 30\%$ at 1.65 eV [25]) caps how far any interface-inclusive screening can go.

11.0 Conclusion

Grid screening under interface-inclusive guardrails finds a 23.84%-efficient best point in the stated CsPbI3/CBTS box and 20.82% at routine defect levels, with every load-bearing assumption priced by computed sweep: single-crystal-grade defects, a low gap, a work-function-tuned front contact, and post-hoc affinity adoption. The same campaign prices the routine-defect thickness optimum, shows the interface dose–response to be regime-dependent (dominant at the defect floor, nearly flat at routine defects), replaces an analytic resistance derate with scripted parasitic sweeps, and validates the doping by Mott–Schottky extraction. We recommend guardrailed reporting (interface status *with* dose level and defect level, defect floors *with* cross-sections, unit-system declaration, R_s basis, contact work functions, affinity choices) as a publication standard for SCAPS studies.

Data availability

All definition files (csPbI3-CBTS-r7.def with interfaces, csPbI3-CBTS-noIF.def without, plus work-function and cross-section variants), sweep receipts (roundA.jsonl, roundB.jsonl, char.jsonl, s0.verify.json, s1_dose.json, s4_sig.json, s5_morris.jsonl, s7_r3.json,

s8_champ38.json), raw curves (champion_JV.json, noIF_JV.json), analysis outputs and figure scripts are archived in the public repository <https://github.com/touhidsiddiqueeraj-bit/cspbi3-cbts-guardrailed> (inventory in Table 14; directories defs/, outputs/, receipts/, scripts/, manuscript/). Reference PDFs under sources/ are copyrighted and listed in REFERENCES.md instead.

Author contributions

The authors designed the screening study, built the SCAPS definition files, executed the simulation campaign, performed the audit and sensitivity analyses, and wrote the manuscript.

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Appendix A: two scripting traps and a verification rule

Unit trap. SCAPS definition files store SI units (metres, m^{-3} , m^2/Vs) but script `set` commands expect practical units (μm , cm^{-3} , cm^2/Vs)—verified by dose–response: `set layer2.NA 1e15` reproduces the file baseline (stored 10^{21} m^{-3}) exactly, while SI-valued inputs (e.g. 10^{22}) inject a $10^6\times$ overdose and yield degenerate ghost cells (V_{oc} 1.47–1.53 V, FF 91–92%, up to 27%). Practical flags that catch them: V_{oc} above the E_g -implied ceiling for the modelled gap, or FF above Green’s ideal-diode value at the computed V_{oc} .

Structural trap. `set interface<n>.IFdefect<k>.capture_cross_section` does not adjust σ in place; it appears to restructure the interface defect block, and in our hands it neutralised the addressed interfaces entirely (one interface: 23.8565%; both interfaces: 25.1494%, essentially the no-interface value). The definition-file route (rewriting `sigma_nleft/.../pright`) reproduced the $N \times 10$ result to four decimals, which the `set-key` route could not.

Rule. For any parameter that changes a defect population, verify the applied value with a `get` of the same field in the same script and record the returned number in the run receipt; treat a result that moves efficiency in the wrong direction as a failed application, not a physical finding.

Appendix B: run inventory

All simulation batches are inventoried in Table 14.

Table 14: Simulation batches archived in the repository (run counts; worker retries noted where they occurred).

Batch	Configurations	Receipt
Baseline validation	4 (one worker-crash retry)	baseline_results.json
Unit diagnostics	6 + 4 retries	diag_units.json
Round A grid	96 (3 non-converged)	roundA.jsonl
Round B refine	36 (3 non-converged)	roundB.jsonl
Characterisation + audit + T	17 + 6 re-runs	char.jsonl
Set-command verification	5	s0_verify.json
Interface dose + Rs/Rsh + WF	14	s1_dose.json
$N_t \times \sigma$ joint	3	s4_sig.json + s4b_sig.json
Morris screen	56 (0 errors)	s5_morris.jsonl
σ verify + χ_{CBTS} + Nt1e15 grid + realistic + noIF-JV	16 (1 worker retry)	s6_rev2.json + noIF_JV.json
Affinity adoption (χ_{TiO2}) + adopted-base anchors	12	s7_r3.json + s8_champ38.json
Morris bootstrap intervals	5000 resamples	s5_morris.ci.json
Routine-defect $\sigma \times 1$ baseline	1	s9_nt15base.json
Attainable gap + E_t sweep	6	s10_r4.json
Sobol-on-surrogate	RF, $N = 4096$ Saltelli	sobol_surrogate.json